

KARTHEEK MUKKAVILLI

Austin, TX | 832-752-3951 | kartheek.mukkavilli@gmail.com | linkedin.com/in/kartheek-mukkavilli | github.com/mkskart | U.S. Citizen

EDUCATION

University of Texas at Austin

Austin, TX

B.S. Electrical & Computer Engineering | GPA: 3.95/4.0 | Expected May 2028

Coursework: Embedded Systems · Digital Logic Design · Digital Signal Processing · Discrete Mathematics · Software Design · Circuit Theory

Elements of Computing Certificate (CS) | Business Foundations Minor

TECHNICAL SKILLS

Languages: Python · C · C++ · Java · TypeScript · SQL · Assembly

Frameworks/Tools: React · FastAPI · Spring Boot · Node.js · Docker · SQLAlchemy · AWS IoT Core · Git

Concepts: REST APIs · Microservices · OAuth 2.0 · CI/CD · RAG Pipelines · Agentic Systems

Certifications: PCAP (Python Associate) · PCEP (Python Entry) · Amazon Junior Software Developer

EXPERIENCE

Human Centered Robotics Laboratory — UT Austin

Austin, TX

Research Assistant

Jan 2026 – Present

- Integrated a PyTorch Transformer to predict 16-DOF bionic hand motion from glove telemetry, achieving 1ms sensor-to-actuation latency.
- Refined a modular PID motor control API for 10 joints, improving tracking from unreliable to consistently stable real-time replication.

National Oilwell Varco

Houston, TX

Embedded Systems Intern

Jun – Aug 2025

- Replaced a \$20K/site legacy PLC system with a custom nRF9160 board under \$250 (98% cost reduction), deployed across 50 field units.
- Engineered Zephyr RTOS C firmware for LTE-M/GNSS over a 40-node LoRa mesh to AWS IoT Core, writing a safe-shutdown protocol to prevent LTE tower rejection on power cycle.
- Automated TLS certificate provisioning via Python + AT%CMNG, cutting per-unit onboarding from hours to under one minute.

J.P. Morgan Chase

Houston, TX

Software Engineering Intern

Jun – Jul 2022

- Built a Spring Boot/Cassandra microservice designed for a 2M+ record production keyspace with dynamic filtering and sorting.
- Optimized SQL query logic reducing analytical latency by 40%; delivered a ReactJS + AG Grid metadata dashboard for 200+ engineers.

PROJECTS

J.A.R.V.I.S. | Python · FastAPI · Ollama · Whisper.cpp · ChromaDB · Porcupine

- Built a fully local voice AI on RPi 5 — Porcupine wake-word, Whisper.cpp STT, Ollama LLM, Mem0 + ChromaDB RAG memory, and tools for Gmail, Calendar, browser automation, and Telegram.

Smart Scheduler | Python · FastAPI · React · TypeScript · Docker · OAuth 2.0

- Architected a greedy deadline/priority scheduling backend with 120+ Pytest unit tests (85% coverage); containerized via Docker Compose with Google OAuth 2.0.

QuantClaw — Algorithmic Trading Engine | Python · Alpaca API · Isolation Forest · Groq API

- Genetic algorithm breeds 50 VWAP+RSI strategy variants over 50 generations daily; 70% win rate over 250 backtested trades, +20% net return, +\$160 on \$800 paper account in one month.

LiteWing — Drone Flight Controller | C++ · FreeRTOS · ESP32

- 1 kHz FreeRTOS PID loop with measured 1.0ms period / 45µs jitter (Saleae); ±2.5° attitude hold over 60-second flight test via complementary filter IMU fusion.

LEADERSHIP

UT IEEE RAS Engineer | TSA Co-Founder: 3× National Finals | 4× Gold PVSA Medalist | SAMPADA Violin Certified · ABRSM Grade 5